

THE UNIVERSITY OF AZAD JAMMU & KASHMIR, MUZAFFARABAD



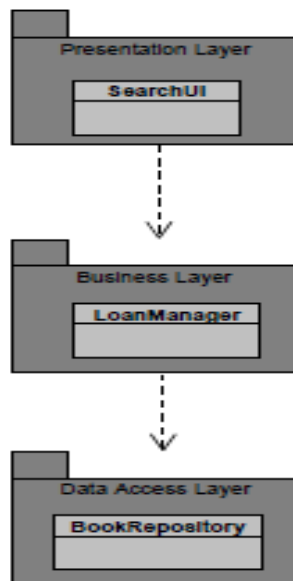
COURSE TITLE	SDA
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DEPARTMENT OF SOFTWARE ENGINEERING

SD&A

Lab 9

Diagram



Scenario: A user wants to Borrow a Book.

1. **Presentation Layer:** The user clicks the "Borrow" button on the screen. The LoanUI captures the BookID and UserID and sends a request to the LoanManager in the layer below.
2. **Business Logic Layer:** The LoanManager receives the request. It first checks if the UserID has any unpaid fines (logic check). If clear, it calculates the due date (e.g., +14 days) and asks the Data Access Layer to save the new loan record.
3. **Data Access Layer:** The LoanRepository executes an SQL command (e.g., INSERT INTO Loans...) to update the database. It confirms the save was successful and returns a "Success" signal back up the chain.